

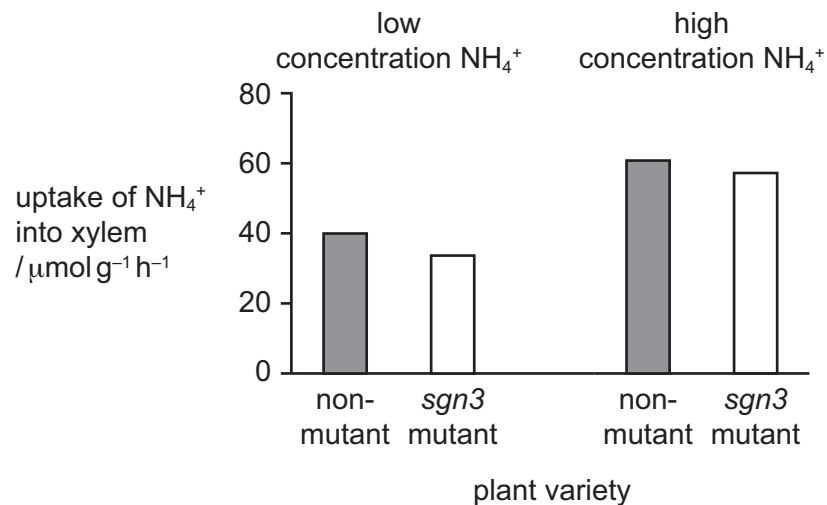
26 Ammonium ions, NH_4^+ , can enter the xylem of plant roots by two pathways.

- In the apoplast pathway, ammonium ions move through cell walls until this pathway is blocked by the Casparian strip. Ammonium ions then enter the cytoplasm of root cells.
- In the symplast pathway, ammonium ions move through the cytoplasm of root cells.

A scientist measured the uptake of ammonium ions into the xylem of *Arabidopsis thaliana*.

The *sgn3* mutant of *A. thaliana* does not have a Casparian strip. In the *sgn3* mutant, ammonium ions can enter the xylem without entering the cytoplasm of root cells.

The bar charts show the scientist's results.



Which conclusion is correct?

- A** Fewer ammonium ions enter the xylem when they have to move through the cytoplasm of root cells.
- B** More ammonium ions enter the xylem at low soil concentrations of ammonium ions.
- C** The Casparian strip acts as a barrier to reduce the movement of ammonium ions into the xylem.
- D** The loss of the Casparian strip has little effect on the movement of ammonium ions into the xylem.